

MA-SQLGrid: A Robust Multi-Agent Framework for Text-to-SQL in Power Grid Databases

Liu Bijing ^{1,2}, Sun Chenglong ^{1,2} and Yang Yong ^{1,2,*}

¹ NARI Group Corporation (State Grid Electric Power Research Institute), Nanjing 211106, Jiangsu Province, China;

² Beijing Kedong Electric Power Control System Co., Ltd., Beijing 100080, China

* Correspondence: Yang Yong; email address to be provided before submission

Featured Application

MA-SQLGrid is an experimental read-only decision-support framework for auditable natural-language access to structured power-grid records. Generated SQL and returned source rows require human inspection before consequential use; no operational control, protection, or maintenance decision has been validated.

Abstract

Natural-language access to power-grid databases requires traceable schema grounding, database-enforced read-only execution, and explicit handling of missing evidence. This paper presents MA-SQLGrid, a five-role framework coordinated through an append-only blackboard and deterministic adjudicator. Its revised executor uses SQLite read-only URI mode, `query_only`, a mutation-denying authorizer, disabled extensions, and time, opcode, row, table, and column limits. Counterfactual evidence is fail-closed: when named states are required, incomplete coverage makes a candidate ineligible. Earlier GridDB and BIRD experiments evaluate frozen prompting workflows, not autonomous multi-agent generation. A deterministic no-generation descriptive re-execution applies three selectors to one shared precomputed evidence collection for eight historical candidates on each of 180 GridDB questions. Within each v3 run, all boards were sealed before the execution path directly loaded raw gold; however, the frozen pre-run regression suite had accessed gold-derived v2 outcomes for the same items. The result is therefore descriptive rather than confirmatory evidence from previously inaccessible outcomes. All 5760 candidate–state attempts, including 332 failures, were retained, and two reproductions were identical. Fixed-order accuracy was 80/180 (0.4444). Validation-only ranking achieved 100/180 (0.5556), whereas complete reference-free metamorphic selection achieved 101/180 (0.5611). The selectors differed on only Q039: the M3 nullable-schema-extension witness rejected a wildcard projection and retained an explicit-column projection. This post-audit example is a narrow projection-stability trace, not evidence of a general counterfactual or multi-agent gain. Both selectors covered 180/180 and harmed two first-candidate successes. Reversed tie order yielded 117–118/180, demonstrating substantial rule dependence. Inherited BIRD Mini-Dev results remain a non-grid portability comparison with unequal method call counts. The evidence supports a reproducible safety and coordination architecture plus bounded descriptive behavior on a historical pool, not a five-role end-to-end gain, autonomous-agent benefit, qualified power-grid semantic validation, deployment safety, or universal robustness.

Keywords: text-to-SQL; multi-agent systems; power-grid databases; schema grounding; execution validation; counterfactual testing; reproducibility

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1. Introduction

Power-grid organizations maintain relational data about assets, locations, topology, sensor readings, work orders, technicians, and maintenance activity. SQL provides precise access to these records, but it requires knowledge of table names, join keys, identifiers, units, aggregation conventions, and local coding rules. Text-to-SQL systems aim to reduce this access barrier by translating a natural-language request into an executable query. In an engineering workflow, however, a fluent query is not sufficient evidence of correctness. A query can execute while selecting the wrong status field, omitting a required filter, using an unintended join path, or returning the expected number of columns with the wrong content.

Cross-database benchmarks such as Spider and BIRD have made schema generalization and realistic database content central evaluation concerns [1,2]. Schema-aware models, constrained decoding, context selection, decomposition, and execution guidance address different failure surfaces [3–7]. These methods also expose a design problem: generation steps are frequently bundled together. A system may change schema serialization, value presentation, domain normalization, structural hints, candidate count, and repair policy simultaneously. An observed score difference then cannot be assigned cleanly to schema linking, reasoning, or validation.

The original MA-SQLGrid concept addresses this problem through five specialized roles. A Query Analyst interprets the request, a Schema Cartographer maps it to the database, an SQL Synthesizer packages candidate queries, a Validation Engine checks safety and execution, and a Counterfactual Critic records evidence from named database states. A separate deterministic adjudicator resolves the resulting evidence. This separation makes the trace inspectable: a reviewer can distinguish what the analyst inferred, which schema elements were retained, which candidates were proposed, which candidates executed, and why a final candidate was selected or rejected.

The architecture must nevertheless be separated from the evidence used to evaluate it. The inherited GridDB factorial study was not a multi-agent execution. It used one model generation per question and prompt cell, followed by deterministic parsing, read-only validation, execution, and offline scoring. The BIRD study likewise compared four frozen prompting procedures rather than five interacting agents. These assets can motivate modules, supply baselines, and test replay interfaces, but they cannot be renamed as results of the new coordination core.

This distinction leads to three questions. First, can role-specific handoffs be implemented with a database-enforced, gold-isolated, fail-closed boundary? Second, what do the completed experiments establish about context packages, presented values, public-database transfer, and constructed-state agreement? Third, when generation is held fixed at eight already frozen candidates per question, how do fixed-order, validation-only, and complete metamorphic selectors behave under the same evidence-collection operations? The third question estimates historical-pool offline selection only; schema grounding is retained as trace evidence and no five-role end-to-end, new-generation, dialogue, or deployment effect is estimated.

The paper makes four contributions. First, it defines a testable five-role coordination core, a database-enforced SQLite executor, and a deterministic adjudicator whose required counterfactual coverage fails closed. Second, it conservatively integrates a 1440-prediction GridDB factorial experiment, a 700-call component study, a 25,920-row multi-state ledger, and a 5000-call BIRD comparison while retaining their original identities and evidence

classes. Third, it reports a hash-frozen offline selection study over the existing eight-slot pool using three query-blind metamorphic witnesses with distinct database hashes; all three methods choose only after the same shared 32-attempt evidence collection per question is complete. Fourth, it reports negative and sensitivity evidence: the full condition changes one Q039 wildcard-to-explicit-column decision relative to validation-only, while tie order materially changes the result.

No Spider or WikiSQL experiment is reported, no unsupported large power-grid question–SQL corpus is asserted, and the new offline result is not relabeled as autonomous multi-agent generation. Qualified domain review and a matched new-generation experiment remain future gates.

2. Related Work

2.1. Text-to-SQL Evaluation and Schema Generalization

Text-to-SQL systems map a natural-language question and a database schema to a query. Spider established a cross-domain setting with previously unseen databases, drawing attention to table selection, joins, grouping, nesting, and ordering [1]. BIRD extends public evaluation toward larger and more content-rich databases [2]. A benchmark result depends on the exact split, evaluator, database engine, execution boundary, and model configuration. We therefore report only the completed BIRD Mini-Dev protocol and make no claim about public datasets that were discussed but not run.

Exact SQL match and execution evaluation answer different questions. String match can penalize equivalent SQL, whereas result equality can reward nonequivalent queries on an insufficiently discriminating database state. A robustness evaluation should retain predicted SQL, record every failure, and specify the state on which equality was observed. MA-SQLGrid uses strict result equality for inherited snapshot analyses and separately reports constructed-state agreement and bounded metamorphic invariance. Neither auxiliary endpoint is semantic certification.

2.2. Schema Linking, Context Packages, and Output Contracts

Schema linking connects question spans to tables, columns, values, and foreign-key relations [3,8,9]. Full-schema context preserves recall but can increase token cost and distraction. Compact context reduces the search space but risks dropping a required join key or predicate. Value examples can disambiguate coded attributes, while question-derived instructions can identify aggregation, result shape, ordering, or a limit.

The inherited GridDB factorial study isolates two implemented prompt factors: a full-DDL/global-values package versus a compact/domain-grounded package, and absence versus presence of a composite structural/SQL-operation hint. The compact package is not a pure schema-length manipulation because it also changes value presentation and adds corpus-tailored normalization. The hint is not merely formatting because it can include aggregation, grouping, projection, ordering, and limit guidance. We retain these precise labels instead of assigning broader causal interpretations.

Constrained generation and execution guidance reduce invalid candidates. PICARD constrains decoding so inadmissible SQL continuations can be rejected during generation [4]. CHESS combines contextual grounding and SQL synthesis components [7]; other prompt-based methods investigate decomposition and representation choices [5,6,10]. MA-SQLGrid differs at the coordination boundary: its new core accepts externally produced candidates, checks them with deterministic contracts, and records evidence before selection. It does not claim that the current lexical cartographer supersedes learned schema-linking systems.

2.3. Agentic Decomposition and Auditable Control

Agentic decomposition can make intermediate decisions explicit, but naming pipeline stages as agents does not establish a multi-agent benefit. Such evidence requires a comparison that holds the model, data, decoding configuration, candidate count, and physical generation budget constant. Otherwise, a multi-candidate system can improve simply because it receives more samples.

MA-SQLGrid treats an agent as a role with a typed contract rather than assuming that every role must be a separately hosted model. The Analyst and Cartographer can be deterministic or model-assisted; the Synthesizer can receive candidates from a frozen generator; the Validator uses a fixed read-only policy; the Critic consumes named-state evidence; and the Adjudicator is deterministic. This design allows components to be replaced without changing the blackboard schema and prevents gold-guided selection from being hidden behind an opaque debate label.

2.4. Power-Grid Data and External Validity

Power-system datasets combine topology, equipment attributes, measurements, time series, costs, and constraints. RTS-GMLC and SimBench provide reproducible engineering resources that can be transformed into relational structures [11,12]. Their availability does not automatically create a text-to-SQL benchmark: intended projections, units, ordering, ties, and reference SQL require semantic review. The current RTS-GMLC, SimBench, and NERC-derived candidates are consequently machine-adjudicated silver resources, not expert gold.

DKA-SQL is a direct power-grid text-to-SQL precedent [13]. Its published corpus, method, models, and evaluation boundary differ from ours. Because no official implementation was used, the current BIRD protocol is neither a DKA-SQL reproduction nor a head-to-head comparison.

3. Materials and Methods

3.1. Task and Evidence Boundary

For question x_i , read-only database D_i , and introspected schema S_i , a generator produces one or more candidates

$$\mathcal{Y}_i = \{\tilde{y}_{i1}, \dots, \tilde{y}_{ik}\}. \quad (1)$$

Each candidate must satisfy a single-statement read-only contract. The accepted leading form is SELECT or WITH; comments, multiple statements, write operations, schema changes, attachments, and pragmas are rejected. A candidate is eligible for adjudication only when it is safe and executable.

The offline evaluator may later compare the selected query with a reference result. Let $E_i = 1$ when the selected SQL executes and its result equals the frozen reference result under the registered evaluator, and $E_i = 0$ otherwise. Reference SQL, reference results, E_i , and any derivative correctness field are forbidden from all five agents and the Adjudicator. Evaluation occurs only after the blackboard is sealed.

3.2. Data Resources and Provenance

GridDB-Maintenance-v2 v0.1 is a synthetic SQLite case study with eight tables and 98 rows: `asset_types` (6), `assets` (18), `grid_topology` (9), `locations` (8), `maintenance_logs` (8), `sensor_readings` (26), `technicians` (8), and `work_orders` (15). It contains 200 natural-language-SQL records: 20 development records and 180 factorial evaluation records. The evaluation records include 66 easy, 91 medium, and 23 hard items and map to 70 normalized

gold-SQL structural clusters, of which 58 are singletons. The evaluation partition had been visible during project development; it is a controlled finite-set case study, not a sealed production benchmark.

BIRD Mini-Dev contributes 500 public items over 11 SQLite databases. The authorized protocol froze item and method order, prompts, two local model files, evaluator behavior, Python 3.10.11/SQLite 3.40.1, and zero retries. Each backbone made 2500 calls and produced 2000 final predictions. An independent audit re-executed all 4000 predictions and 500 gold queries with zero score mismatches. Two earlier failed attempts, totaling 2476 physical calls, remain immutable incidents and are excluded from scores.

The GridDB reliability release contains 18 states: the canonical snapshot, three insertion permutations, and 14 additional schema-valid stress states. The scorer produced 25,920 prediction-state rows (1440×18). A prediction-blinded automated protocol admitted a 66-question subset for the primary 15-state analysis and retained 114 order-sensitive questions as diagnostics. These constructed states are not operator-certified grid snapshots.

Artifacts are labeled Inherited, Recomputed, New, or Diagnostic. The coordination core is implementation evidence and its replay is Diagnostic. Neither is a New execution-accuracy result.

Table 1 separates resources by domain, evidence class, visibility, and semantic-review status. BIRD is explicitly non-grid. RTS-GMLC and SimBench provide authentic power-system structures, but their question-SQL pairs remain machine-generated silver candidates with zero completed qualified-human reviews and zero sealed items.

Table 1. Data-resource scope and evidence status. Counts describe retained local artifacts; they are not interchangeable benchmark denominators.

Resource	DBs	Tables	Rows	NL-SQL	Evidence class	Visibility and review status
GridDB v0.1	1	8	98	200	Synthetic grid; inherited development case	20 dev + 180 evaluation; development-visible; no documented independent dual-expert review
GridDB suite	state 18 states	8 state-dependent	0 new	0 new	Constructed grid-state diagnostic	Prediction-blinded construction; not operator-certified
RTS-GMLC pilot	1	10	360,530	55	Authentic grid structure; machine silver	AUTO_CANDIDATE; 0 human-reviewed; 0 sealed; source-use notice applies
SimBench pilot	1	8	874	36	Authentic grid structure; machine silver	Development-visible; 0 human-reviewed; 0 sealed; ODbL/DbCL review required before redistribution
BIRD Mini-Dev	11 dataset-specific	dataset-specific	500	Public portability benchmark	non-grid power-system benchmark	Frozen public split; BIRD v1.1 protocol; not evidence of power-grid validity

3.3. Five-Agent Coordination Framework

The evidence chronology is shown in Table 2. Release v3 is classified as a post-audit descriptive re-execution: its execution path sealed decisions before directly loading raw gold, but the frozen regression suite had already read v2 gold-derived outcomes for the same 180 items. Consequently, hashing and within-run sealing provide traceability, not prospective outcome blindness.

MA-SQLGrid comprises five specialist roles coordinated by a deterministic controller. Figure 1 is derived from the implemented source rather than an aspirational free-form

Table 2. Study chronology, visibility, and permitted interpretation.

Study	Data/candidate visibility	Rule or analysis freeze	Outcome access	Permitted interpretation
GridDB factorial	Evaluation partition visible during development	Original protocol before generation; v3 hierarchy is post-review re-analysis	After generation	Inherited finite-corpus prompt-package evidence
Component study	Candidate prompts frozen before calls	Component protocol frozen before calls	After black-board/selection seal	Inherited component evidence; not five-role coordination
Formal-v5 states	States built prediction-blind	Scorer frozen before formal scoring	Gold-relative scoring after state freeze	Constructed-state diagnostic
BIRD v1.1	Public Mini-Dev	Protocol/runtime frozen before 5000 calls	Formal evaluation after calls	Non-grid workflow portability; unequal calls by method
Retrospective replay	All predictions and T0 evidence already existed	Diagnostic logic hash-locked	No gold accuracy computed	Candidate-pool coverage only
R2 descriptive re-execution v3	Eight historical candidates and their prior evaluations already existed	Question-only selection view, three distinct witnesses, construction code, tests, and rules hash-bound in v3; frozen tests had read same-item v2 outcomes	Raw gold directly loaded by the v3 execution path after all 180 boards were sealed	Reproducible descriptive selector behavior on previously evaluated items; no prospective, five-role, or generation-effect claim

agent diagram. The controller is not a sixth reasoning model: it orders calls, applies the registered rule, and seals the trace. Messages receive consecutive sequence numbers, and the complete trace is serialized canonically to a SHA-256 digest.

The **Query Analyst** produces a structured question-only intent record containing lexical tokens, detected aggregations, ordering requirements, and an explicit limit when present. It does not select SQL or see correctness. The **Schema Cartographer** maps lexical intent to tables, columns, and foreign-key edges. The current skeleton uses deterministic overlap and records unmatched tokens; a richer GridDB adapter adds disclosed domain rules and graph expansion.

The **SQL Synthesizer** packages externally produced SQL strings. It contains no model client, performs no hidden API call, canonicalizes whitespace and semicolons, removes exact duplicates without changing first occurrence, and assigns stable identifiers. The **Validation Engine** separates lexical admissibility from the executor trust boundary. The R2 executor opens a frozen database through a URI with `mode=ro&immutable=1`, enables SQLite `query_only`, disables loadable extensions, and installs an authorizer that admits only selection, reads, approved functions, and recursive-query opcodes. Writes, DDL, transactions, attachments, pragmas, metadata enumeration, dangerous file/extension functions, and unauthorized table or column reads are denied. A progress handler enforces registered time and opcode limits, and result fetching enforces a row cap. Every attempt records database and SQL hashes, authorization or resource failure, elapsed time, row count, and result hash; no automatic retry removes a failure. These controls establish database-enforced read-only and bounded execution for the tested SQLite boundary. They do not establish user entitlement, data minimization, process isolation, or operational authorization.

The **Counterfactual Critic** consumes named state results, rejects duplicate state identifiers, and records evaluated, passed, failed, and missing states. When counterfactual evidence is optional, missing evidence remains diagnostic and does not break prior no-counterfactual behavior. When the caller requires counterfactual evidence, coverage must

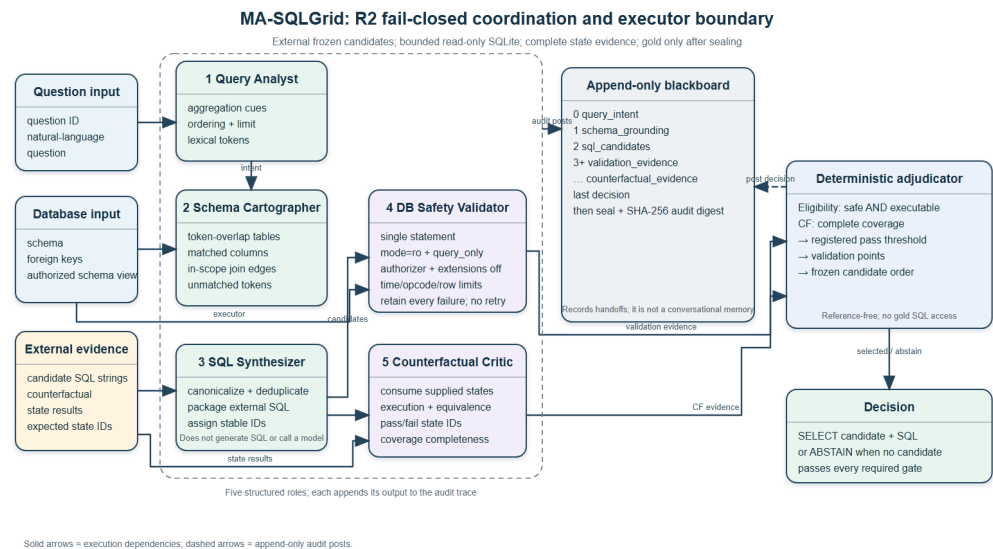


Figure 1. Implemented R2 MA-SQLGrid boundary (code-native SVG and deterministic raster derivative). Candidate SQL is supplied externally. The database validator enforces read-only and resource limits; counterfactual-required selection demands complete named-state coverage; gold SQL remains outside the blackboard until sealing. User authorization and operational approval remain outside this software boundary.

equal the complete registered state set and the pass threshold must be met; otherwise the candidate is ineligible. Thus an incomplete 1/1 record cannot outrank a complete 10/11 record. Existing formal-v5 labels compare predictions with gold and remain forbidden during selection.

3.4. Deterministic Adjudication and Abstention

Eligibility always requires safety and successful execution. The default rule assigns 40 validation points for safety, 40 for execution, 10 for question-derived aggregate-shape conformity, 5 for question-derived ordering conformity, and up to 5 for lexical value hits. In no-counterfactual mode, ties preserve original candidate order. In required-counterfactual mode, incomplete coverage fails closed before ranking and the registered pass threshold is enforced. If no candidate remains eligible, the controller abstains. These are deterministic recorded engineering rules, but v3 does not establish that their design preceded access to derived outcomes for the 180 items. Weight, pass-threshold, and reversed-order results are therefore descriptive sensitivity analyses.

For this paper, robustness is a vector rather than a universal label (Table 3). Only tested components are reported as evidence.

3.5. Inherited GridDB Factorial Design

The paired design crosses context package $c \in \{0, 1\}$ and composite hint $h \in \{0, 1\}$. F00 uses full DDL and global values without the hint; F01 adds the hint; F10 uses compact domain-grounded context without the hint; and F11 combines compact context and the hint. The same 180 questions appear in all four cells for each backbone, yielding 720 Qwen and 720 Granite predictions.

Qwen2.5-Coder-7B Q4_K_M and Granite-3.3-8B Q4_K_M were run at temperature zero with fixed snapshots. Each prompt produced exactly one response. The parser retained the first SQL candidate through the first semicolon; the validator then applied the read-only policy. There was no candidate ranking, multi-agent negotiation, or repair in this experiment.

Table 3. Operational robustness dimensions and their evidence boundaries.

Dimension	Registered operation/end point	Evidence in this paper	Not established
Mutation safety	DB-enforced denial of write/DDL/attach/pragma operations	Adversarial executor tests; unchanged database hash	User authorization or whole-process sandboxing
Resource boundedness	Timeout/opcode and maximum-row termination	Recursive-query and row-overflow tests; retained failure traces	Worst-case bounds for every SQLite feature
Evidence completeness	Complete named-state set required before CF-aware eligibility	Fail-closed tests and complete three-state ledgers	Missing-not-at-random state correction
Metamorphic invariance	Selected result equals T0 under irrelevant-relation, index/rebuild, and nullable-extension witnesses	180-question offline selector result	Value drift, paraphrase, production-state, or semantic robustness
Semantic validity	Agreement with frozen GridDB reference after board sealing	Finite historical-pool accuracy	Qualified human gold or field-operational correctness

Algorithm 1 Gold-isolated deterministic coordination

Require: question x_i , schema S_i , external candidates $\mathcal{Y}_i$, read-only executor, named state evidence

- 1: post structured intent and schema grounding to the append-only blackboard
- 2: canonicalize and deduplicate externally produced candidates
- 3: **for** each candidate $y \in \mathcal{Y}_i$ **do**
- 4: apply single-statement read-only validation
- 5: record execution and available reference-free state evidence
- 6: **end for**
- 7: **if** required named-state coverage is incomplete for a candidate **then**
- 8: mark that candidate ineligible (fail closed)
- 9: **end if**
- 10: **if** no safe executable candidate meets all registered gates **then**
- 11: abstain
- 12: **else**
- 13: apply the fixed validation, state-evidence, coverage, and order rule
- 14: **end if**
- 15: post the decision and seal the blackboard
- 16: load reference results only for offline evaluation

Strict execution equality is primary. A common-target projected-column indicator is a secondary manipulation check. Paired finite-set contrasts use normalized-SQL clusters as dependence proxies, registered sign-flip tests, composition-sensitivity bootstrap intervals, and Holm correction. The intervals describe sensitivity to the observed corpus composition, not population confidence.

3.6. *Component, Multi-State, and Public Protocols*

The separately frozen component study contains 700 scored calls. E1 compares candidate generation with and without presented value evidence. E2 requests up to three candidates and compares a deterministic reference-answer-independent selector with the first candidate. Selection is sealed before gold loading. Rescue, harm, and oracle-at-three are descriptive; oracle-at-three is a gold-only upper-bound diagnostic.

The v5 reliability protocol reproduced all 1440 T0 labels and then executed every prediction over 18 states. The primary 66-question analysis uses a logical-AND endpoint over 15 semantic states. A prediction passes only when it agrees with the reference on every included state. The nine registered contrasts form a separate Holm family; with only 12 structural clusters, all $2^{12} = 4096$ sign assignments are enumerated.

The BIRD protocol compares direct prompting, decomposition, schema selection, and bounded execution repair. Each method has a fixed call pattern and zero retries. Pairwise differences use exact database-cluster sign randomization across 11 databases and Holm correction.

3.7. *Retrospective Offline Coordination Diagnostic*

The replay verifies SHA-256 for two 720-row prediction ledgers and the 25,920-row formal-v5 ledger before reading them. For each question it pools the eight existing outputs from two backbones and four prompt cells, canonicalizes SQL, and removes duplicates. Frozen T0 fields provide execution and metadata-shape evidence. The Critic records counterfactual evidence as unavailable because existing state-agreement labels are gold-relative.

This is a coverage diagnostic. Candidates were generated under different models and prompt packages; the pool is not a deployable, matched multi-agent run. No selected output is compared with gold to produce an accuracy number.

3.8. *Deterministic Descriptive Offline Re-Execution Study v3*

The R2 v3 study reuses exactly eight frozen source slots for each of the 180 GridDB evaluation questions: four Qwen prompt cells followed by four Granite prompt cells. It performs no LLM call and creates no new question or SQL label. Before any method selects, the three methods share one precomputed eight-by-four execution ledger: T0 plus three metamorphic witnesses for every slot. Thus 5760 is the shared physical evidence-collection count, not a separate natural runtime cost assigned to each selector. The fixed-order control then ignores the collected ranking evidence and chooses slot 0; validation-only passes an empty counterfactual mapping and ranks validation evidence only; the full condition requires complete 3/3 reference-free metamorphic invariance. Incomplete coverage or no passing candidate causes abstention.

The three query-blind witnesses have distinct SHA-256 values and are built from T0 without reading questions, prompts, predictions, scores, or gold. M1 adds an irrelevant relation and rows; M2 adds harmless indexes and rebuilds storage; M3 adds nullable probe columns to three business tables. M1 and M2 preserve original-schema denotation. M3 preserves explicitly projected original columns but deliberately allows wildcard projection to fail. Query order is preserved when candidate SQL contains ORDER BY; otherwise rows are compared as a multiset. The selector sees only question text extracted from

frozen prompt ledgers, the schema, eight SQL strings, executor traces, and T0-to-witness equivalence. It does not see gold SQL, answer-shape metadata, correctness, difficulty, feature tags, required literals, or gold-relative formal-v5 labels.

Before v3 execution, a content-bound UTC timestamp, configuration, question-only prompt-derived selection view, two 720-row prediction ledgers, schema, T0, three witness databases, complete witness manifest, exact witness-builder bytes and SHA-256, executor, coordinator, core and release runners, all four regression-test files, and the passing pre-run test record were bound into a 21-file manifest with content SHA-256 8b34bc370451173197ad07b460908537836409d6eb49f303caf46d13d53889e6. The witness manifest independently binds its builder and three database hashes. Within each v3 run, all 180 blackboards and their decisions were sealed and written before that execution path directly loaded raw gold. Each run retains all 1440 source slots and all $180 \times 8 \times 4 = 5760$ shared attempts, including failures, without retry. Thirty regression tests passed, and run v3a and run v3b produced byte-identical selection, evaluation, sensitivity, and summary files. Independent release audit passed mechanical integrity and descriptive traceability but failed the prior-outcome-independence evidence-class gate: the frozen pre-run test suite had opened v2 gold-derived evaluation rows for these same 180 items. V3 must therefore be interpreted as deterministic descriptive re-execution, not as a scientific release independent of prior same-item outcomes. V1 and v2 remain preserved diagnostic predecessors.

Descriptive endpoints are strict execution accuracy over all 180 questions, coverage and abstention, complete metamorphic invariance of the selected candidate, and rescue/harm relative to fixed order. The retained sensitivity matrix crosses three weight policies, invariant-pass thresholds of 1/3, 2/3, and 3/3 while always requiring complete observation, and original versus reversed candidate-order tie breaking. Because same-item outcomes were accessible before v3, this matrix is reported only as descriptive analysis.

4. Results

4.1. Complete GridDB Cell Results

All 1440 scheduled predictions are present, and independent SQLite re-execution reproduced the stored execution verdicts. Table 4 reports strict execution equality.

Table 4. Strict execution equality over 180 GridDB questions per cell.

Backbone	F00 full/no hint	F01 full/hint	F10 compact/no hint	F11 compact/hint
Qwen	0.4222	0.7167	0.4333	0.6000
Granite	0.4278	0.5556	0.4111	0.6000

The common-target projected-column diagnostic is 0.5222 in F00 for both backbones. Qwen obtains 0.9667, 0.5889, and 0.9611 in F01, F10, and F11; Granite obtains 0.8778, 0.5667, and 0.9222. Returning the requested number of columns is therefore easier than returning the intended rows. In Qwen F01, projected-column conformity is 0.9667 while execution equality is 0.7167.

4.2. Factorial Effects and Multiplicity

For Qwen, the composite-hint execution effect is +0.2306, the package-hint interaction is -0.1278, and the compact-package main effect is -0.0528. Granite’s corresponding estimates are +0.1583, +0.0611, and +0.0139. Zero of nine primary execution tests survives Holm correction. Qwen’s hint effect has raw/adjusted $p = 0.01372/0.09604$, its interaction 0.00919/0.07352, and the cross-backbone hint modifier 0.00640/0.05760. No primary factorial execution effect or modifier is therefore described as statistically nonzero.

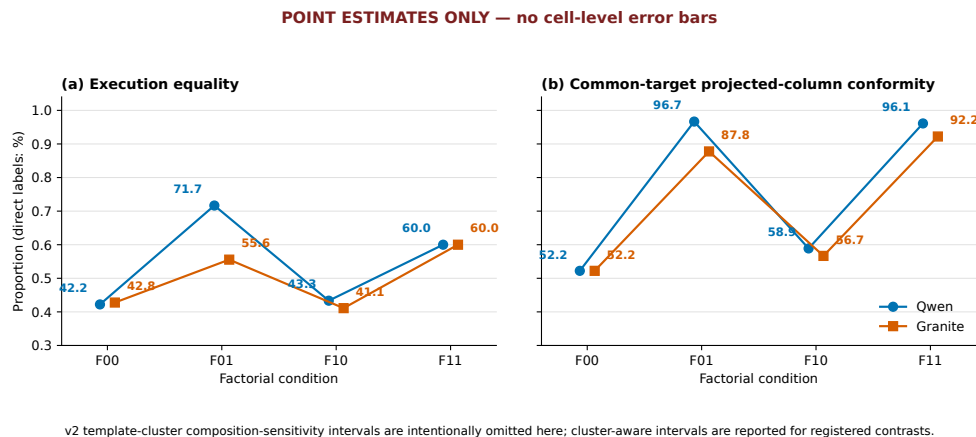


Figure 2. Audited GridDB cell point estimates. Each cell contains all 180 attempts; the figure is descriptive and does not replace registered factorial contrasts.

In the separately corrected secondary family, the common-target hint effects survive: +0.4083 for Qwen (adjusted $p = 0.000090$) and +0.3556 for Granite (adjusted $p = 0.01944$). Because the hint supplies structural information later scored by this endpoint, these are manipulation-check results, not independent semantic evidence.

4.3. Component Results

The component run completed all 700 scored calls. Presented value evidence increased Qwen first-candidate execution equality by +0.1059 over 170 eligible questions (95% composition-sensitivity interval [+0.0282, +0.2013], adjusted $p = 0.0310$). Granite’s estimate was zero with an interval spanning zero. The cross-backbone modifier did not meet the registered rule, so the Qwen result is not replicated across backbones.

The deterministic selector changed 24 Qwen and 36 Granite choices. Relative to the first candidate, it rescued eight and harmed one Qwen question, and rescued ten while harming none for Granite. Execution effects were +0.0389 and +0.0556; neither survived its two-test Holm family. Oracle-at-three values of 0.6389 and 0.4667 are gold-only diagnostics and are not deployable selector results.

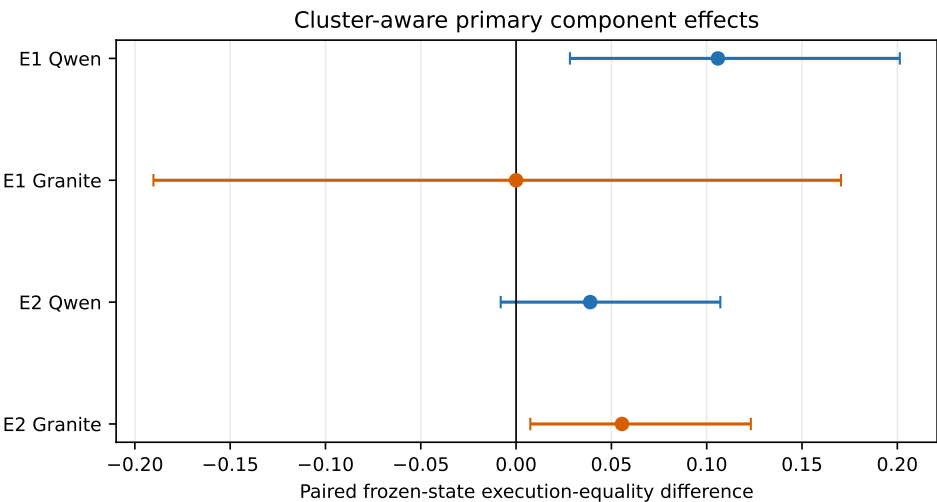


Figure 3. Prospectively frozen component effects. Intervals describe composition sensitivity; the registered decision also requires multiplicity-adjusted evidence.

4.4. Multi-State Reliability Results

The v5 scorer produced all 25,920 registered rows, and its T0 slice reproduced 1440/1440 snapshot labels. Across the 66-question automatic subset, 15-state logical-AND rates range from 0.6212 (Granite F01) to 0.8182 (Granite F00). All nine Holm-adjusted values equal 1.0000 under exact cluster-sign enumeration. No context, hint, interaction, or cross-backbone multi-state effect meets the declared rule. These rates characterize agreement over constructed witness states, not population accuracy or human-certified equivalence.

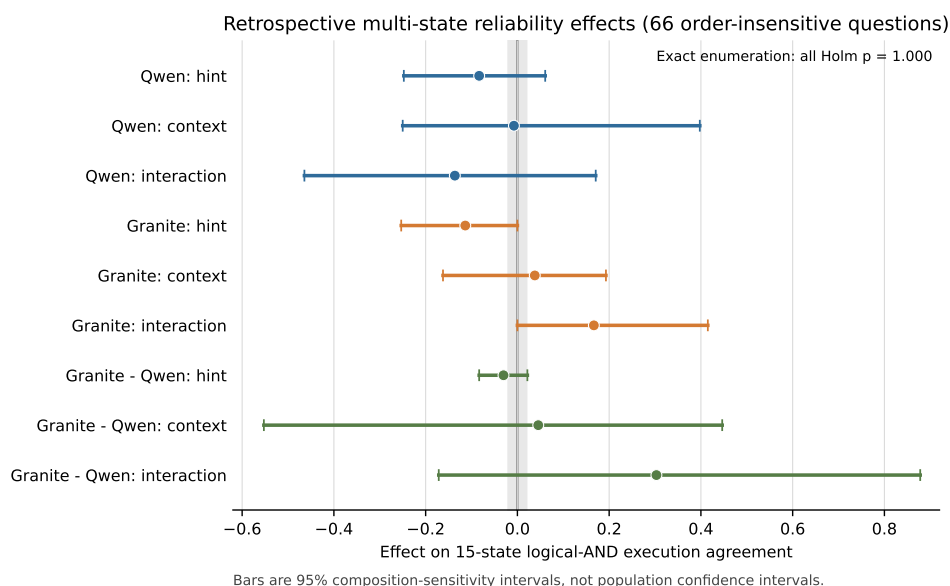


Figure 4. GridDB multi-state logical-AND agreement for the prediction-blinded automatic subset. The endpoint is a constructed-state robustness diagnostic.

4.5. Public BIRD Results

Table 5 reports all eight cells. BIRD Mini-Dev is a public non-grid portability benchmark and does not validate power-grid semantics. B0–B2 use one physical generation call per item; B3 uses two, so the inherited comparison estimates frozen workflow efficacy under unequal call counts rather than a call-matched repair effect. Each backbone completed 2500 calls, 2000 final predictions, zero retries, and zero final-ledger omissions; latency and token totals were not promoted into the retained summary and are therefore reported as unavailable rather than reconstructed.

After Holm adjustment, Qwen decomposition is 0.076 below direct prompting (adjusted $p = 0.0430$), and schema selection is 0.092 above decomposition (adjusted $p = 0.0117$). No Granite contrast survives correction. Different method orders across the two model snapshots argue against a universal prompting recipe. The two failed incident runs totaling 2476 physical calls remain excluded and immutable; they are not retries contributing to these denominators.

4.6. Retrospective Replay Coverage

The replay audited all 180 GridDB questions without a generation call. At least two distinct frozen SQL candidates were present for 173 questions. After safety checks and binding to consistent T0 execution evidence, 172 questions had at least two eligible candidates and entered retrospective adjudication. Seven failed closed because only one unique SQL remained, and one failed because fewer than two candidates were eligible.

Unique candidate counts ranged from one to eight: 1 (7 questions), 2 (19), 3 (15), 4 (18), 5 (27), 6 (23), 7 (45), and 8 (26). These counts show that frozen assets can populate the

Table 5. Inherited BIRD Mini-Dev v1.1 results (non-grid portability evidence; 11 database clusters). Intervals are database-cluster composition-sensitivity intervals.

Backbone	Method	Correct/500	Accuracy	Interval	Calls/item	Failures
Qwen	B0 direct	189/500	0.378	[0.286, 0.470]	1 0	final omissions
Qwen	B1 decomposition	151/500	0.302	[0.203, 0.401]	1 0	final omissions
Qwen	B2 schema selection	197/500	0.394	[0.279, 0.511]	1 0	final omissions
Qwen	B3 execution repair	174/500	0.348	[0.250, 0.452]	2 0	final omissions
Granite	B0 direct	102/500	0.204	[0.125, 0.284]	1 0	final omissions
Granite	B1 decomposition	105/500	0.210	[0.131, 0.294]	1 0	final omissions
Granite	B2 schema selection	101/500	0.202	[0.137, 0.273]	1 0	final omissions
Granite	B3 execution repair	118/500	0.236	[0.159, 0.318]	2 0	final omissions

Table 6. Retrospective offline coordination diagnostic coverage. Counts are not accuracy and do not estimate multi-agent improvement.

Coverage gate	Questions
Audited	180
At least two unique frozen SQL candidates	173
At least two eligible candidates	172
Retrospectively adjudicated	172
Failed closed: one unique candidate	7
Failed closed: fewer than two eligible candidates	1
Reference-free counterfactual evidence available	0

coordination interfaces for most questions. They do not show that retrospectively selected SQL is correct. No retrospective accuracy, execution gain, rescue rate, or multi-agent superiority is reported.

4.7. Descriptive Offline Coordination-Selection Results

The deterministic descriptive re-execution retained 5760 candidate-state attempts; 332 were non-executable or authorization/resource/SQL failures and remained in the all-attempt ledger. Every question nevertheless had at least one eligible candidate, so both adjudicating selectors covered 180/180 questions and abstained on 0/180. Table 7 reports the recorded rules.

Table 7. New v3 offline selection over one historical eight-slot pool per question. All methods choose after the same shared precomputed 32-attempt evidence collection. Accuracy is finite-corpus strict execution equality after all blackboards were sealed; this is not a new-generation or five-role end-to-end experiment.

Selector	Correct/180	Accuracy	Covered	Abstained	Invariant	Rescue/harm
Fixed order, equal budget	80/180	0.4444	180	0	177/180	0/0
Validation rank, equal budget, no CF	100/180	0.5556	180	0	179/180	22/2
Complete metamorphic coordination	101/180	0.5611	180	0	180/180	23/2

Validation-only ranking added 20 net correct items relative to fixed order through 22 rescues and 2 harms. Complete metamorphic coordination changed the selected candidate on only Q039, yielding one further correct item (101/180 versus 100/180), one additional rescue, no additional harm, and no coverage or abstention change. On Q039, validation-only retained a SELECT * candidate; M3 added a nullable column to work_orders, so complete metamorphic selection rejected that wildcard projection and selected an explicit work_order_id, scheduled_date projection. This 1/180 observation is a narrow nullable-schema-extension projection-stability case. It is descriptive and too small, dataset-specific, and rule-dependent to support a general counterfactual benefit or five-role end-to-end claim. Schema grounding is recorded in the trace but does not affect the selector in this study.

All recorded weight policies with original-order tie breaking produced 101/180 for the full condition, and invariant-pass thresholds from 1/3 through 3/3 produced the same selection because every selected winner had complete 3/3 evidence. Reversing candidate-order ties produced 117/180 under the default and intent-emphasis weights and 118/180 under validation-equal weights. These more favorable values are descriptive sensitivity results, not confirmatory alternatives. The 16–17-item shift shows that the apparent offline behavior is materially rule/order-dependent. The result supports auditability and exposes a design risk; it does not establish a generally robust or superior selector.

5. Discussion

5.1. What the Evidence Supports

The combined evidence supports four bounded findings. First, explicit structural and SQL-operation instructions strongly alter projected-column adherence, but surface conformity does not establish correct execution. Second, presented value evidence can matter for one frozen model snapshot while showing no effect for another. Third, candidate selection and prompting are backbone-dependent: the component selector did not meet its registered efficacy rule, and the descriptively best BIRD method differed between Qwen and Granite. Fourth, on the historical eight-slot pool, validation ranking improved the observed finite-set score relative to fixed order, but complete metamorphic evidence added only 1/180 and tie-order sensitivity was large.

These findings justify separating intent, schema grounding, synthesis, validation, and state evidence. They do not establish that five roles are superior to one call or that newly generated multi-agent candidates would reproduce the offline gain. The architecture's demonstrated value is methodological and safety-bounded: typed contracts, database-enforced eligibility, complete-evidence gates, deterministic selection, abstention, incident retention, and a sealed gold boundary make later comparisons reproducible and falsifiable.

5.2. Interpreting the Retrospective Replay

The replay asks whether inherited ledgers contain enough distinct executable candidates to exercise the coordination interfaces. For 172 of 180 questions, they do. This reduces implementation risk and identifies eight fail-closed cases that require a registered fallback or a new matched generation policy.

The replay cannot estimate a coordination effect. Its pool mixes two backbones and four prompt packages, so generation condition and diversity are confounded. It also lacks reference-free counterfactual equivalence evidence. Post hoc gold scoring of selected outputs would remain exploratory and cannot substitute for a prospective matched comparison.

5.3. Interpreting the Descriptive Re-Execution

The R2 study hash-binds the selection view, rules, code, tests, and state inputs and verifies the software fact that each v3 execution path directly loaded raw gold only after all blackboards were sealed. This does not establish outcome blindness: its frozen test suite had accessed gold-derived v2 evaluation rows for the same items before v3. The study is therefore a deterministic descriptive re-execution over a historical candidate pool. Equal candidate budget means that each selector received the same eight pre-existing slots; it does not mean equal historical model calls between those source conditions, and it does not turn those candidates into contemporaneous agent outputs.

The small contrast between validation-only and complete metamorphic selection is scientifically important. V3 uses three different query-blind database hashes and exercises an irrelevant-relation witness, an index/storage witness, and a nullable schema-extension witness. These are three operator families rather than independent samples from a deployment population. The full gate changes only Q039, specifically because M3 exposes wildcard-projection arity while preserving explicit-column projections. An earlier v1 diagnostic used three same-hash insertion files and is retained as an excluded incident; v2 is retained as a provenance-incomplete diagnostic predecessor; none of those predecessor outcomes enter the v3 release claim.

The reversed-order sensitivity exposes a second limitation: many candidates share the same coarse safety and intent scores. The deterministic tie rule consequently selects a backbone/prompt slot without semantic evidence. Reporting only the 0.5556 default or choosing the 0.6556 sensitivity after outcome access would be misleading. A later rule-development study requires development-only calibration, richer reference-free evidence, and a genuinely untouched evaluation resource.

5.4. Power-Grid Use and Reproducibility

Database-enforced read-only execution, resource limits, syntactic validity, and result-shape conformity are necessary safeguards, not operational certification. A safely executed query can still return the wrong assets, omit time bounds, misinterpret codes, or mishandle ties. The executor's optional table/column policy is a testable authorization hook, not evidence that a utility user's permissions were configured. Multi-state witnesses reduce accidental snapshot equality but cannot cover all distributions or schema changes. Gen-

erated SQL and returned source rows require authorized human inspection before any maintenance, protection, dispatch, or asset-management decision.

Negative results are part of the contribution. Zero of nine primary factorial execution tests survives correction. The deterministic component selector meets its efficacy rule for neither snapshot. No registered multi-state effect survives correction. Granite and Qwen do not share the same best BIRD method. Immutable ledgers, direct re-execution, registered test families, portable manifests, and explicit incidents preserve these outcomes rather than collapsing them into a success narrative.

6. Limitations and Future Work

The primary domain corpus is one synthetic database with eight tables, 98 rows, and 180 evaluation questions. Its evaluation partition was visible during development. The 70 structural groups are dependence proxies, and 58 are singletons. Results cannot be generalized to production schemas, permissions, missing-data patterns, units, access-control policies, or workloads.

Only two quantized local snapshots were evaluated. Model family, size, quantization, serving software, and serialization may interact. The compact package bundles schema reduction, values, normalization, and serialization; the hint bundles projection, aggregation, grouping, ordering, and limit rules. The inherited factorial experiment identifies package effects rather than contributions of individual internal rules.

The five-role core and database executor are implemented and pass 30 unit/adversarial/regression tests, but the system has not completed a new-generation experiment on an evaluation resource whose outcomes were previously inaccessible to the study process. Its Analyst and Cartographer are deterministic skeletons, its Synthesizer packages external candidates, and its adjudication weights have not been independently optimized. The R2 offline study descriptively re-executes selection over historical candidates and previously evaluated items; it does not estimate generation, communication, latency, token, or autonomous-agent effects. Existing formal-v5 agreement labels are gold-relative and intentionally excluded from selection.

RTS-GMLC, SimBench, and NERC question-SQL assets remain machine-adjudicated silver data. Qualified review is required for intended projections, units, ordering, tie handling, and result granularity before external accuracy is reported.

Future work should execute a new hash-locked, call-matched comparison of four contemporaneous generation conditions: a single direct candidate; staged question/schema handoffs with one candidate; a fixed multi-candidate budget with validation and adjudication but no counterfactual evidence; and the same candidate pool with a pre-registered multi-family reference-free state suite. Model snapshot, question order, database state, decoding, candidate count, physical calls, and evaluator must be held constant. The protocol must pre-register execution accuracy, valid SQL, abstention, tokens, latency, failure taxonomy, clustered inference, multiplicity, and incident retention. The candidate-order rule must be chosen on development-only evidence before the untouched evaluation set is opened.

Further work can replace lexical grounding with a trained schema linker, add a low-confidence full-schema fallback, compare deterministic adjudication with budget-matched voting, and evaluate risk-coverage calibration. Qualified power-system/database reviewers must independently assess domain semantics; LLM or author-assisted triage cannot be relabeled as independent expert gold. Every change should receive a new protocol identity rather than modifying a completed run.

7. Conclusions

MA-SQLGrid reframes power-grid text-to-SQL as an auditable coordination problem. Five specialist roles separate question analysis, schema mapping, candidate packaging, safety and execution validation, and counterfactual criticism. A deterministic controller records their handoffs, selects only eligible candidates, and abstains when required evidence is incomplete. The R2 executor enforces read-only and resource limits at the SQLite boundary rather than relying on lexical filtering alone.

The inherited experiments provide a substantial but bounded evidence base. The 1440-prediction GridDB study shows strong projected-column instruction uptake but no primary execution effect surviving Holm correction. The 700-call component study supports presented values for one Qwen snapshot, not Granite, and does not validate its selector under the registered rule. The 25,920-row state study finds no corrected factorial agreement effect. The 5000-call BIRD protocol supplies a non-grid public comparison whose best method depends on the backbone and whose methods use unequal call counts. The retrospective replay confirms candidate coverage for 172 of 180 questions without accuracy claims.

The new offline study adds one bounded result: validation-only reached 100/180 and complete metamorphic selection reached 101/180 compared with 80/180 for fixed order, while reversed tie order changed sensitivity results to 117–118/180. The 1/180 full-versus-validation increment, rule dependence, historical mixed-source pool, development-visible GridDB, and absent qualified semantic review prevent a general multi-agent superiority or power-grid deployment claim. The original title is retained as the framework identity; the word robust refers only to the explicitly tested safety, bounded-execution, complete-evidence, and bounded metamorphic mechanisms.

Supplementary Materials: A manuscript-bound editor/reviewer archive has been prepared with hash-bound prompts, predictions, scores, frozen protocols, incident records, independent re-execution audits, statistical tables, figure sources, BIRD v1.1 ledgers, the five-role core, database executor, adversarial tests, and both offline-study reproductions. The public repository at <https://github.com/gaoxingkele/ma-sqlgrid> must be synchronized and tagged to the manuscript archive before submission. Restricted materials remain governed by the Data Availability Statement.

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all-attempt ledgers, both independent offline reproductions, code, tests, generated tables and figures, and retained incident records.

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